

APEX — Device Classes

APEX — Adaptive Payload EXchange

Status: Draft | Version: 0.6.0 | Scope: Registry of device classes (traffic types)

1. Overview

This document is the **registry** of APEX device classes. Each device class corresponds to a `traffic_type` value in the APEX V0 outer header (see [APEX — Core §3.1.1](#)) and is fully specified in its own companion document. The core spec is class-agnostic; the per-class docs define both the **Host role** and the **Device role** ([APEX — Core §1](#)) — including the messages each side sends, state machines, and any class-specific timing.

Class assignment happens during discovery: the device declares its desired class via `device_class_req` in `DEVICE_INFO`, and the host accepts or rejects via `CONFIG_REPLY` ([Core §3.2](#)). Once accepted, frames carrying `traffic_type` equal to that class value are routed to the class-specific handler on both sides.

2. Registry

<code>traffic_type</code>	Name	Description	Spec
0	CONFIG	Core control traffic: discovery, configuration, and the implicit heartbeat. Not a device class — defined by the core spec.	APEX — Core
1	ACTIVATION	Devices that progress through enable and trigger phases.	APEX Device Class — Activation
2	ANALOG_HMI	Analog HMI: control packets (CRSF / MAVLink 2) in, CVBS video out.	APEX Device Class — Analog HMI

3	WAYFINDING	Devices that produce directional cues — "point me there" updates for the operator, from any source.	APEX Device Class — Wayfinding
4	REPEATER	Radio frequency repeater	(TBD)
5	USB_FS_HUB	Full-Speed USB passthrough traffic.	(TBD)

3. Reserved & Custom Traffic Types

`traffic_type` values not enumerated above are reserved for future device classes. Vendor- or platform-specific extensions should be coordinated through this registry rather than reusing CONFIG message IDs or values from another class.

Open: Whether to formally allocate a high range of `traffic_type` values (e.g., `0x80-0xFF`) for vendor-custom classes, analogous to the prior spec's `0x80-0xFF` MSG_ID range.

4. Authoring a New Class Spec

A device-class spec should cover, at minimum:

1. **`traffic_type` value** assigned in the registry above.
2. **Roles** — what the Host and Device each do in this class.
3. **Message set** — the inner payload format(s) carried under this `traffic_type`, distinguished by an in-class message ID where applicable.
4. **State machines** — for both host and device, if any.
5. **Timing** — heartbeat cadence beyond the core implicit heartbeat, command-response timeouts, etc.
6. **Failure handling** specific to the class.